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Sunwoo Kim

PhD Student — HLS compilers and LLM agents for chip design

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Education

Cornell University, PhD in Computer Engineering (Advisor: Prof. Zhiru Zhang) Aug 2025 – present

- Coursework: Advanced Compilers; High-Level Digital Design Automation; AI for Systems Engineering

Seoul National University, BS in Electrical and Computer Engineering, cum laude Mar 2019 – Feb 2025

Publications

From Pragmas to Partners: A Symbiotic Evolution of Agentic High-Level Synthesis *(link)* ICCAD 2026
Niansong Zhang, **Sunwoo Kim**, Andrew Butt, Shreesha Srinath, Zhiru Zhang

G^{3}SA: A GPU-Accelerated Gold Standard Genomics Library for End-to-End Sequence Alignment *(link)* ICS 2025
Yeejoo Han*, **Sunwoo Kim***, Seongyeon Park, Jinho Lee *(*equal contribution)*

Experience

PhD Student Researcher, Computer Systems Laboratory, Cornell University Aug 2025 – present

- Led and mentored ~30 MS, MEng, and undergraduate students in the Zhang group on MiniTPU and Allo

Head Teaching Assistant, ECE 4950/6950, Cornell University Jan 2026 – present

- Developed the hardware design and programming labs for a new course on AI hardware

Research Intern, Accelerated Intelligent Systems Lab, Seoul National University Jan 2024 – Jun 2025

- Co-led kernel design and indexing optimization for G^{3}SA, a multi-GPU library covering all NGS alignment stages

Projects

MiniTPU Sep 2025 – present

- Full-stack TPU: VLIW RTL, Triton-based compiler, CUDA-like runtime, and GPT-2/Qwen2.5 chatbot serving
- ASIC implementation in 45 nm and 7 nm (in progress); FPGA prototypes on ZCU104 and Alveo U280

Agentic HLS (Allo) Sep 2025 – present

- Extended Allo, an MLIR-based HLS for domain-specific accelerators, to ASIC; TPU in ~15× less code than RTL
- Integration with ACT to automatically generate software and hardware from a single MLIR architectural spec
- LLM agent harness for IP customization, compiler extension, and post-training setup studies (in progress)

Service

- Artifact Evaluation Committee: ASPLOS 2026, MLSys 2026, SOSP 2026, MICRO 2026

Honors

- Fulbright Graduate Study Award in STEM, $40k (declined), 2024
- SNU Semiconductor Specialization Scholarship, $11k, 2023
- Korea Presidential Science Scholarship, full tuition and stipend, 2019–2024

Skills

Languages: C, C++, CUDA, Python, Bash, Verilog, SystemVerilog

Tools: MLIR, Allo, PyTorch, Vitis HLS, Vivado, cocotb, SPICE, git